

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I Athavan K(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Athavan K

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Question 1 (10 Marks)

A company's virtual machines are consuming excessive CPU and memory resources. Analyze the virtualization architecture and propose methods to optimize resource allocation.

1. Introduction

Excessive consumption of CPU and memory by virtual machines (VMs) is one of the most common operational problems faced in virtualized data centers. It typically arises not because physical resources are inherently insufficient, but because the virtualization architecture has not been configured, monitored, or governed correctly. To resolve the issue in a structured way, it is necessary to first understand the layered architecture through which physical resources are converted into virtual resources, then identify where inefficiencies are creeping in, and finally apply targeted optimization techniques at the appropriate layer.

2. Virtualization Architecture Overview

A typical server virtualization architecture consists of the following layers:

- The underlying compute, memory, storage, and network resources of the physical server (host).Physical Hardware Layer:
- A Type-1 (bare-metal) hypervisor such as VMware ESXi, Microsoft Hyper-V, or KVM runs directly on the hardware and is responsible for creating, scheduling, and isolating VMs. It abstracts physical CPU cores, RAM, and I/O devices into virtual equivalents (vCPU, vRAM, virtual NICs/disks).Hypervisor (VMM) Layer:
- Each VM runs its own guest operating system and applications, believing it has dedicated access to hardware, while the hypervisor actually multiplexes physical resources among all VMs.Virtual Machine Layer:
- Tools such as vCenter, System Center Virtual Machine Manager, or OpenStack that provide resource pooling, scheduling policies, monitoring, and automation across a cluster of hosts.Management/Orchestration Layer:

Excessive CPU/memory consumption occurs when the hypervisor's scheduling and allocation mechanisms are not aligned with actual workload demand, causing resource contention among VMs sharing the same physical host.

3. Root Causes of Excessive Resource Consumption

- VMs are allocated far more vCPU/vRAM than their workload requires, wasting capacity and increasing scheduling overhead.Over-provisioning:
- The hypervisor allows more virtual resources to be assigned than physically exist (a normal practice), but without reservations/limits this leads to contention when many VMs peak simultaneously.Resource Overcommitment without Control:
- Without continuous visibility into per-VM utilization, runaway processes or memory leaks inside a guest OS go undetected.Lack of Monitoring:
- VMs with complementary or conflicting workload patterns are not balanced correctly across hosts (e.g., all CPU-intensive VMs placed on the same host).Poor VM Placement:
- A VM's vCPUs and memory span multiple NUMA nodes, forcing costly remote-memory access and increasing effective CPU cycles consumed.NUMA Misalignment:
- Static resource allocation cannot adapt to variable workload demand, so administrators over-allocate "just in case," which paradoxically increases contention cluster-wide.Absence of Elastic Scaling:

4. Proposed Optimization Methods

- Use utilization history (CPU ready time, memory balloon/swap statistics) to reduce vCPU/vRAM allocations to match actual workload, freeing capacity for other VMs.Right-Sizing VMs:
- Group VMs into resource pools and apply shares (relative priority), reservations (guaranteed minimum), and limits (hard cap) so that critical workloads are protected while non-critical ones cannot starve the host.Resource Pools with Shares, Reservations, and Limits:
- Enable cluster-level automated load balancing (e.g., VMware DRS, Hyper-V Dynamic Optimization) so VMs are live-migrated to less-loaded hosts automatically when contention is detected.Dynamic Resource Scheduling (DRS) / Load Balancing:

- Employ transparent page sharing, ballooning, and compression so idle or duplicate memory pages are reclaimed from VMs and returned to the free pool without impacting performance. Memory Reclamation Techniques:
- Configure large VMs to fit within a single NUMA node (vNUMA alignment) to avoid remote memory access latency and reduce effective CPU consumption. NUMA-Aware Scheduling:
- For cloud-hosted or private-cloud VMs, use horizontal/vertical auto-scaling policies driven by CPU/memory thresholds so capacity grows only when genuinely needed and shrinks when demand falls. Auto-Scaling and Elastic Provisioning:
- Deploy monitoring/APM tools (e.g., vRealize Operations, Prometheus/Grafana, Azure Monitor) with alert thresholds on CPU ready %, memory ballooning, and swap activity to catch problems before they degrade the cluster. Continuous Performance Monitoring:
- Use affinity/anti-affinity rules so that VMs with complementary usage patterns (e.g., a CPU-bound and an I/O-bound workload) are co-located, while similarly peak-heavy VMs are spread apart. Workload-Aware VM Placement:
- Conduct regular reviews of allocated versus consumed resources and hold business units accountable via chargeback/showback so over-provisioning requests are minimized at the source. Periodic Capacity Planning and Chargeback Audits:

5. Conclusion

Excessive CPU and memory consumption in a virtualized environment is rarely a hardware shortage problem; it is usually an architecture, configuration, and governance problem. By combining right-sizing, share/reservation/limit-based resource pools, automated load balancing (DRS), memory reclamation, NUMA-aware placement, and continuous monitoring, an organization can restore predictable performance, improve consolidation ratios, and defer unnecessary hardware expenditure.

Question 2 (10 Marks)

Explain how VM sprawl affects cloud performance and recommend administrative policies to prevent it.

1. What is VM Sprawl?

VM sprawl refers to the uncontrolled proliferation of virtual machines within an organization's infrastructure, where VMs are created quickly (due to the ease of provisioning in a virtualized/cloud environment) but are never properly tracked, maintained, or decommissioned. Over time, a large number of idle, duplicate, forgotten, or under-utilized VMs accumulate, silently consuming compute, storage, licensing, and management resources.

2. Causes of VM Sprawl

- Spinning up a new VM takes minutes, so teams create VMs for short-term testing or demos and forget to delete them. Ease of provisioning:
- Anyone with access can create a VM without justification or an expiry policy. Absence of an approval/lifecycle workflow:
- VMs are not tagged with an owner, project, or purpose, making it impossible to know if they are still needed. Lack of ownership tracking:
- Projects end but their VMs are never cleaned up because no one is responsible for teardown. No decommissioning process:

3. Impact of VM Sprawl on Cloud Performance

- Idle and forgotten VMs continue to reserve CPU, memory, and storage capacity, reducing the resources available to legitimate production workloads and increasing contention across the cluster. Resource Contention:
- The hypervisor and cluster scheduler must track and periodically evaluate a much larger inventory of VMs, increasing management overhead and slowing down DRS/load-balancing decisions. Degraded Scheduling Efficiency:

- Every additional VM consumes storage for its virtual disks and snapshots, and is included in backup/replication jobs, lengthening backup windows and increasing storage costs.Storage and Backup Overhead:
- Unpatched, forgotten VMs become weak points that are not monitored for vulnerabilities, increasing the risk of compromise, which can also indirectly affect the performance and integrity of the shared infrastructure.Increased Security Attack Surface:
- Guest OS and application licenses tied to unused VMs inflate costs without adding business value, reducing budget available for scaling genuinely needed capacity.Higher Licensing and Operational Costs:
- Sprawl distorts utilization data, making it difficult for administrators to accurately forecast when additional physical hardware is genuinely required.Complicated Capacity Planning:

4. Recommended Administrative Policies to Prevent VM Sprawl

- Require a request-and-approval process for every new VM, capturing owner, business justification, expected lifetime, and cost center.Formal VM Provisioning Workflow:
- Enforce tags for owner, project, environment (dev/test/prod), and expiry date on every VM at creation time.Mandatory Tagging/Metadata Standards:
- Configure test/dev VMs with an automatic expiration date after which they are powered off and, following a grace period, deleted unless renewed.Automated Expiry and Decommissioning:
- Schedule periodic (e.g., monthly/quarterly) reviews using monitoring tools to identify idle, zombie, or orphaned VMs and flag them for review.Regular Utilization Audits:
- Bill or report VM costs back to the owning department so teams have a financial incentive to decommission unneeded VMs.Chargeback / Showback Model:
- Limit VM creation rights to trained administrators or through a self-service portal governed by quotas per team/project.Role-Based Access Control (RBAC) on Provisioning:
- Provide pre-approved templates with right-sized CPU/memory so ad-hoc, oversized VMs are discouraged.Standardized VM Templates and Sizing Guidelines:
- Maintain a Configuration Management Database that automatically reconciles actual running VMs against approved records to surface unauthorized or forgotten instances.Centralized Inventory and CMDB Integration:

5. Conclusion

VM sprawl silently erodes cloud performance by consuming compute, storage, and management capacity that could otherwise serve active workloads, while also increasing cost and security risk. Preventing it requires a combination of governance (approval workflows, tagging, RBAC), automation (expiry policies, monitoring-driven audits), and financial accountability (chargeback/showback), ensuring that the virtualized environment remains lean, well-utilized, and performant.